

DEVELOPMENT REPORT

Introduction

The Linux Simulator was developed with the aim of practising Python programming and applying the knowledge acquired throughout my studies. The project began as a simple idea to create a system similar to an operating system running in a terminal environment, but it gradually evolved as new features were added.

Initial Planning

At the beginning of development, the main objective was to create an interactive environment that simulated some of the functions found in real operating systems. The intention was not to reproduce a complete Linux system, but rather to build a project that would allow me to practise programming logic, code organisation, and Object-Oriented Programming.

The initial ideas included:

- Menu system.
- Option-based navigation.
- Code organisation through functions.
- Terminal-based interface.

Early Versions

The first versions were simple and contained only the basic features. The initial focus was to ensure that the program worked correctly before adding more advanced functionality.

During this stage, several tests were carried out to verify the operation of the menus, user input handling, and navigation between options.

Project Evolution

As development progressed, new features were implemented:

- Calendar.
- Brasília clock.
- Navigation system.
- Calculator.
- Interactive games.
- Folder management.

- **Browser simulation.**

Each new feature required planning, testing, and corrections to ensure that the system remained organised and functional.

Main Challenges

During development, several challenges were encountered:

- **Organising the code as the project grew.**
- **Integrating different features into a single system.**
- **Correcting logical errors.**
- **Properly structuring classes and functions.**
- **Maintaining code readability.**

These challenges contributed to both learning and the improvement of programming skills.

Knowledge and Skills Gained

The development of the Linux Simulator helped improve knowledge in:

- **Python.**
- **Object-Oriented Programming.**
- **Conditional statements.**
- **Loop structures.**
- **Project organisation.**
- **Problem-solving.**
- **Debugging.**

In addition to technical knowledge, the project also helped develop persistence and planning skills.

Final Outcome

At the end of development, a simulator containing several integrated features within a single system was created. The project served as an important learning experience and represents the progression of the knowledge acquired during programming studies.

Future Improvements

Some planned improvements for future versions include:

- **Better module organisation.**
- **A more advanced interface.**
- **New features.**
- **Improved error handling.**

- **Expansion of the simulator's available resources.**